COMP396 weekly Meeting Minutes

(week 6)

Date & Time: Friday, March 6th, 2026, 14:00–18:00

Location: Harold Cohen Library, Upper Ground Floor, Group Study Table 2

Note Taker: Zhang Olivia (ZO)

Attendees: Zhang Olivia (ZO), Zhao Saria (ZS), Xue Ella (XE), Wong Haw (WH)

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1. Agenda

- Review the final implementation of the TeamStrategy code against module requirements.
- Synthesize the sub-strategy logic and capital allocation documentation for the CA2 report.
- Confirm the parameter optimization methodology (Grid Search).
- Prepare for the next supervisor meeting.

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2. Discussion & Decisions

2.1. Task Review and Report Progress

Our team had successfully completed all tasks assigned last week. The report content is now

significantly advanced:

- **Detailed Strategy Logic:** ZO completed the "Detailed Logic of the Three Sub-strategies" section. To ensure accuracy, XE assists ZO to finalize the "Final Strategy and Parameter Configuration Table" for all assets. This table explicitly lists the strategy type, specific logic, key parameters, and entry/exit conditions for each of the ten series .
- **Capital Allocation:** WH completed the "Capital Allocation Scheme" section and made the allocation table, assigning 14% to each of the five Trend assets (70% total) and a 6% to each of the five Mean-Reversion/Short assets (30% total).
- **Optimization Proposal:** XE proposed the use of Grid Search for systematic parameter optimization. Our team discussed this and confirmed it as our primary method to refine thresholds and lookback periods before the final submission.

2.2. Code Development Summary (team2.py)

ZS presented the completed TeamStrategy class, which serves as our unified "single class" solution. Based on our review of the code file, our team summarized ZS's implementation as follows:

- **Architecture and Compliance:** The code defines a single TeamStrategy class inheriting from backtrader.bt.Strategy. It strictly follows the requirement to avoid "result-oriented programming" by utilizing structural logic applicable across Part 2 and potentially Part 3 data.
- **Parameter Management:** A params dictionary is defined at the class level to pass allocation variables, such as trend_allocation=0.70, ensuring compatibility with the backtester.
- **Active Strategies:** Successfully replaced "Buy and Hold" with mathematically complex sub-strategies, including Price Breakout, MA Crossovers, Z-Score Mean Reversion, and RSI-based Short-selling .
- **Position Sizing:** Our team implemented an overspend_guard and target value logic to maintain per-asset (14% or 6%) based on dynamic portfolio value.
- **Performance Output:** Preliminary backtesting on Part 2 data yields strong results: a return of 21.4%, an Aggregate PD Ratio of 2.716, and an ‘Activity’ level of 96.9%, all of which align with high-scoring thresholds.

2.3. Code Audit and Compliance Check

Our team conducted a formal audit of the code against the "Code Requirements" document:

- **Time Constraint:** Our team verified that the code logic is efficient and will execute well within the **40-minute** limit.
- **External Packages:** Our team confirmed that only standard PyPI packages (Pandas, NumPy, Statsmodels) are used, ensuring the code is portable.
- **Part 3 Readiness:** The strategy is designed to be instantiated by a main.py script for

anonymous testing on Part 3 data.

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3. Task Allocation (E-Series)

Given the **ACFI342 midterm exam on March 16th**, our team have adjusted our schedule to allow for revision.

ID	Task Description	Owner	Due Date
E1	ACFI342 Midterm Exam Revision (No project tasks).	All Members	——
E2	Attend Supervisor meeting (after 16:00) to obtain feedback on CA2 progress.	All Members	Mar 13
E3	Refine the final report and team2.py code based on supervisor feedback and Grid Search results.	All Members	Mar 13 (18:00)

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4. Supervisor Questions

- Data Naming Consistency: Our current code maps parameters using d._name (e.g., 'series_1'). Our team need to confirm if the Part 3 data naming convention will remain identical to Part 2 to ensure our asset-mapping logic functions correctly.

- Scoring Metric Presentation: Aside from the core metrics (Return, PD Ratio, and Activity), are we strictly required to explicitly present "Taking Positions" and "Best Performance on an Individual Series" in our code?

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5. Next Meeting Arrangement

- **Date & Time:** Friday, March 13th, 2026, 18:00–20:00 (Following supervisor meeting)
- **Location:** Harold Cohen Library, Upper Ground Floor, Group Study Table 2
- **Key Agenda:**
    - Review supervisor feedback and implement necessary changes.
    - Prepare the final file for CA2 submission.